CSUN | COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

ECE 528/L

Robotics and Embedded Systems with Lab

Homework 2

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Section I: Review Questions

Create your own GitHub repository for this homework assignment. Then, provide your answers to the following questions in a README file in your repository. Code statements must be written in C and you must use hexadecimal values for bitmasks. You are encouraged to use external references.

1. (5 pts) What is a hardware timer? What kinds of tasks do embedded systems commonly use hardware timers for?

2. (5 pts) What is a clock signal used for in embedded systems? List the primary clock sources available on the MSP432P401R microcontroller and briefly describe each clock source.

3. (5 pts) Consider a 16-bit timer (e.g. Timer_A configured in up mode). If it counts upward, what is the maximum count value before it rolls over? Show your work.

4. (5 pts) If a 24-bit timer counts upward from zero at 10 MHz, what is its overflow period? Show your work.

5. (5 pts) Describe the SysTick timer on the MSP432P401R microcontroller. Provide two example applications for SysTick in embedded systems. Which clock sources can the SysTick timer use and how can they be selected?

6. (5 pts) How many registers does the SysTick timer have on the MSP432P401R microcontroller and what is the purpose of each? Name them and provide a short description per register.

7. (5 pts) Describe the Timer_A peripheral on the MSP432P401R microcontroller. List the main operating modes of Timer_A and briefly explain each.

8. (10 pts) Briefly state the purpose of the following Timer_A registers.
    - TAxCTL
    - TAxCCTLn
    - TAxR
    - TAxCCRn
    - TaxIV
    - TAxEX0

9. (5 pts) What is PWM? Provide at least three example applications that use PWM in embedded systems.

10. (5 pts) How many Timer_A modules are available and how many PWM outputs can each module provide?

11. (25 pts) Describe the steps needed to initialize P2.4 as a PWM output driven by Timer_A0 CCR1 (TA0.1). Use the Timer_A2_PWM.c driver provided in Lab 2 (Motor Control) as a reference.

    Include C code such that:
    - The P2.4 pin is configured to use its primary module function (Timer_A output)
    - SMCLK is selected as the clock source for Timer_A
    - The timer clock is divided by 2
    - Up/down mode is selected
    - The output mode is configured as Toggle / Reset by setting the appropriate bits of the OUTMOD field in the CCTL[1] register
    - The PWM signal has a period of 20 ms and a duty cycle of 50%

12. (20 pts) Describe the steps needed to configure Timer_A0 to generate a periodic interrupt every 2 ms with an interrupt priority level of 2. Note that Timer_A0 has an interrupt number of 8. Use the Timer_A0_Interrupt.c driver provided in Lab 2 (Motor Control) as a reference.

    Include C code such that:
    - SMCLK is selected as the clock source for Timer_A with a prescaler value of 1 (which means the ID field will be 00b)
    - The TAIDEX field in the TAxEX0 register has a value of 000b (divide by 1)
    - Up mode is selected